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(54) **PWM CAPTURE FUNCTION FOR EVENT ANALYSIS**

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G01R 31/56 (2020.01)

(52) **U.S. Cl.**
CPC **G01R 31/56** (2020.01)

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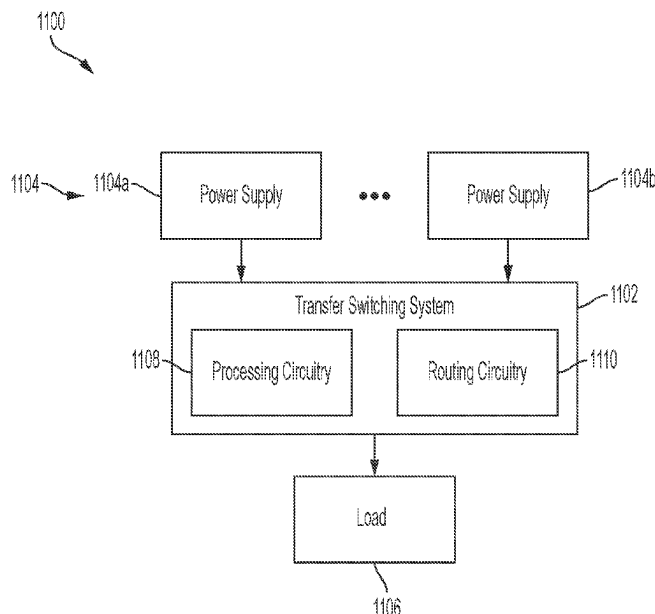
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(57) **ABSTRACT**

According to one example, an electrical device is provided comprising an input configured to receive power from at least one power supply, an output configured to provide the power to at least one load, a first memory configured to store data temporarily, a second memory, the second memory being non-volatile memory, and at least one control device configured to acquire data indicative of at least one parameter of the power responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components, and store the acquired data in the second memory responsive to determining that the fault condition exists.

21 Claims, 13 Drawing Sheets



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(58) **Field of Classification Search**

CPC H02M 1/00; H02M 1/0003; G05B 2219/31034; H02J 3/001

See application file for complete search history.

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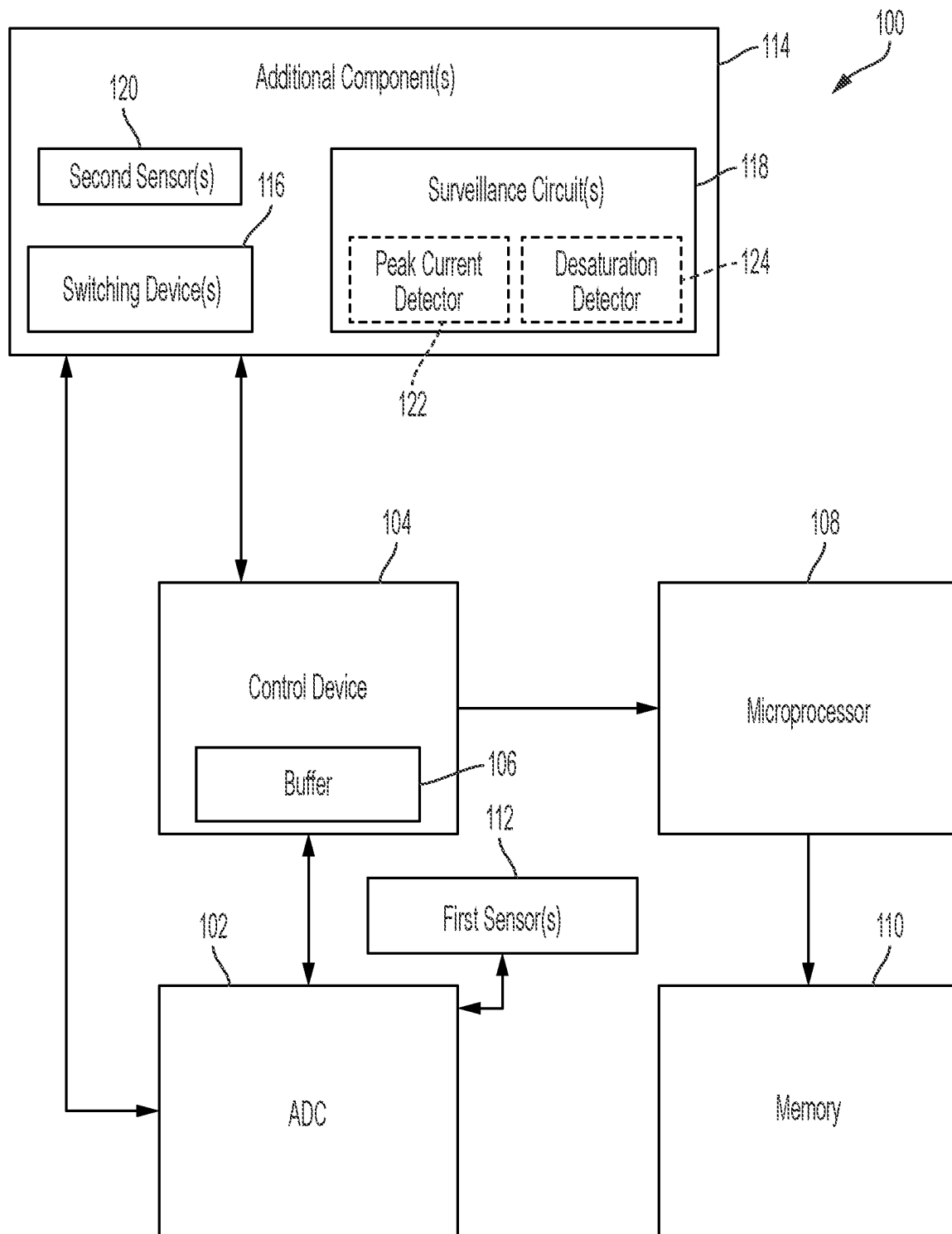


FIG. 1

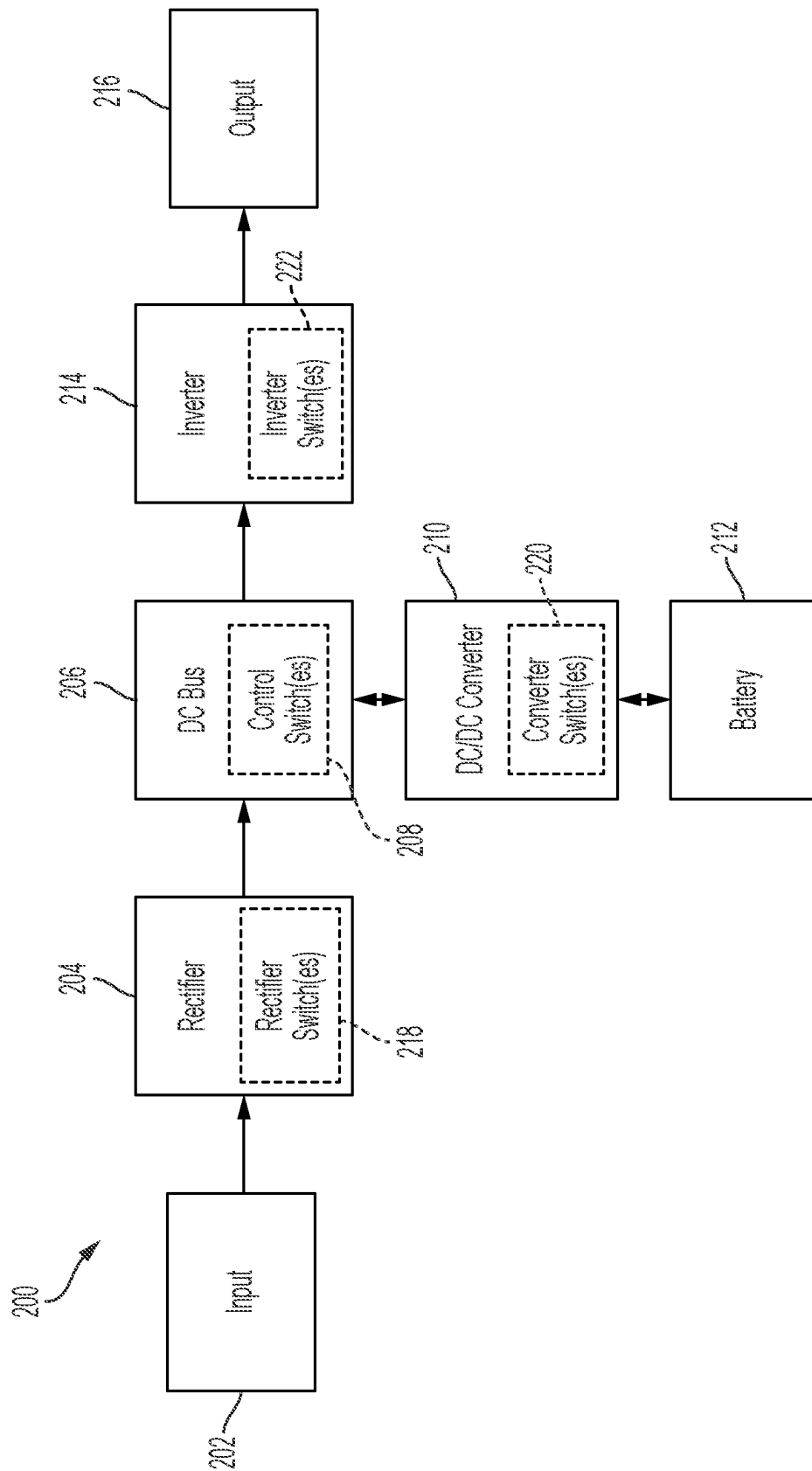


FIG. 2

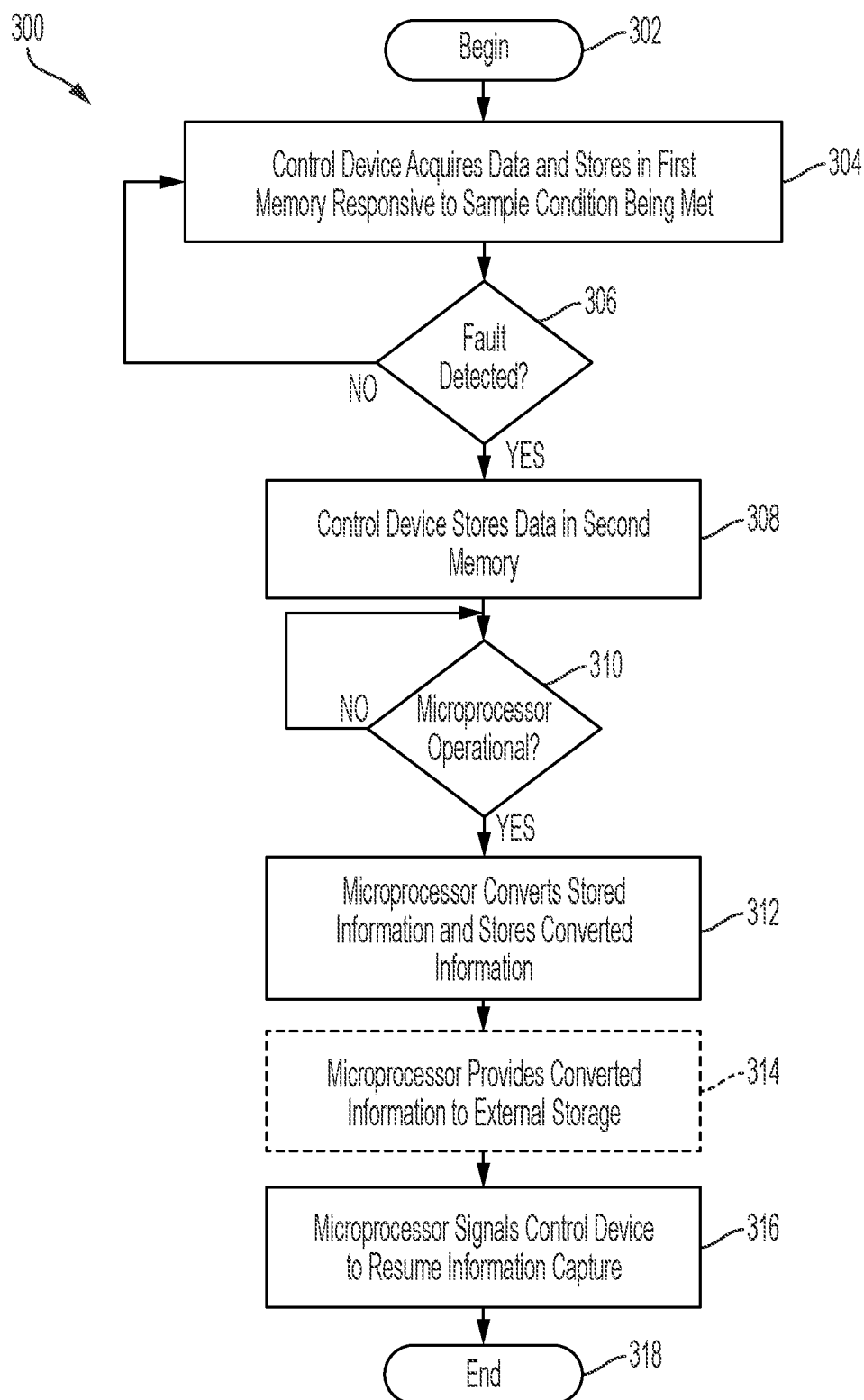


FIG. 3

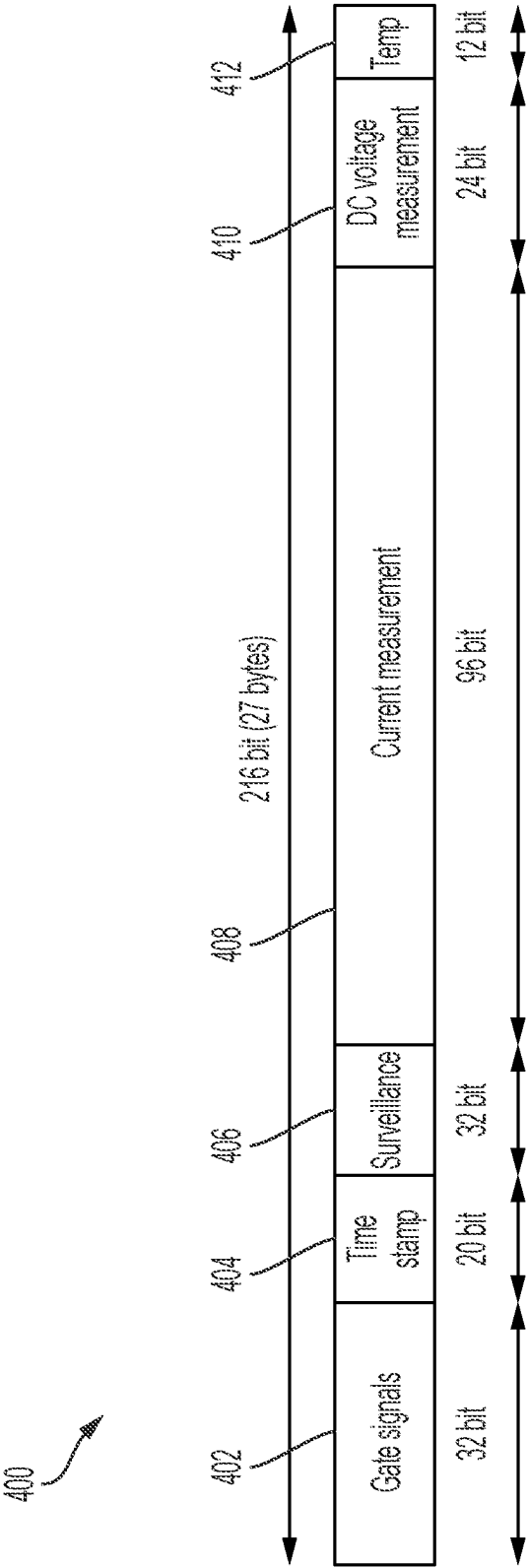


FIG. 4

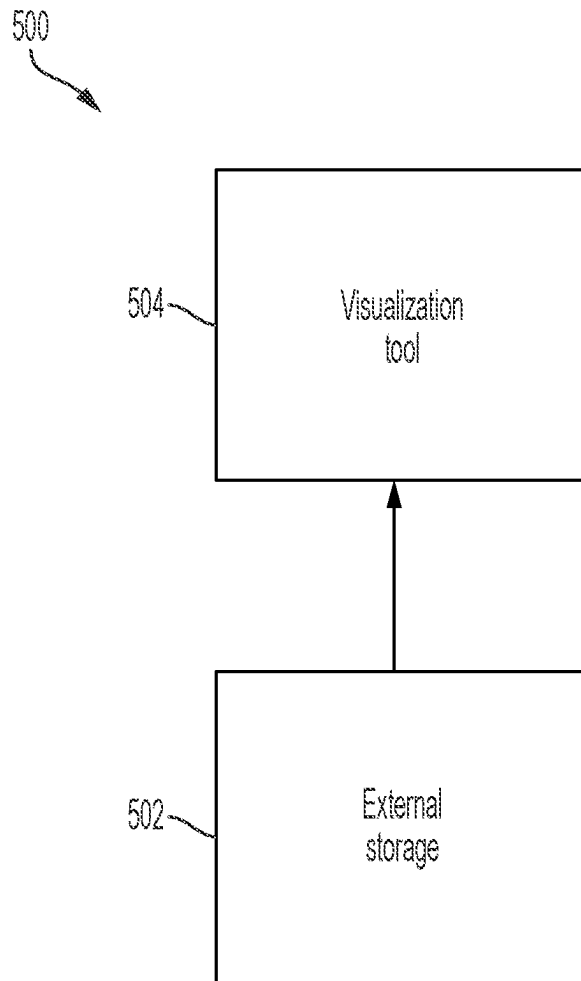


FIG. 5

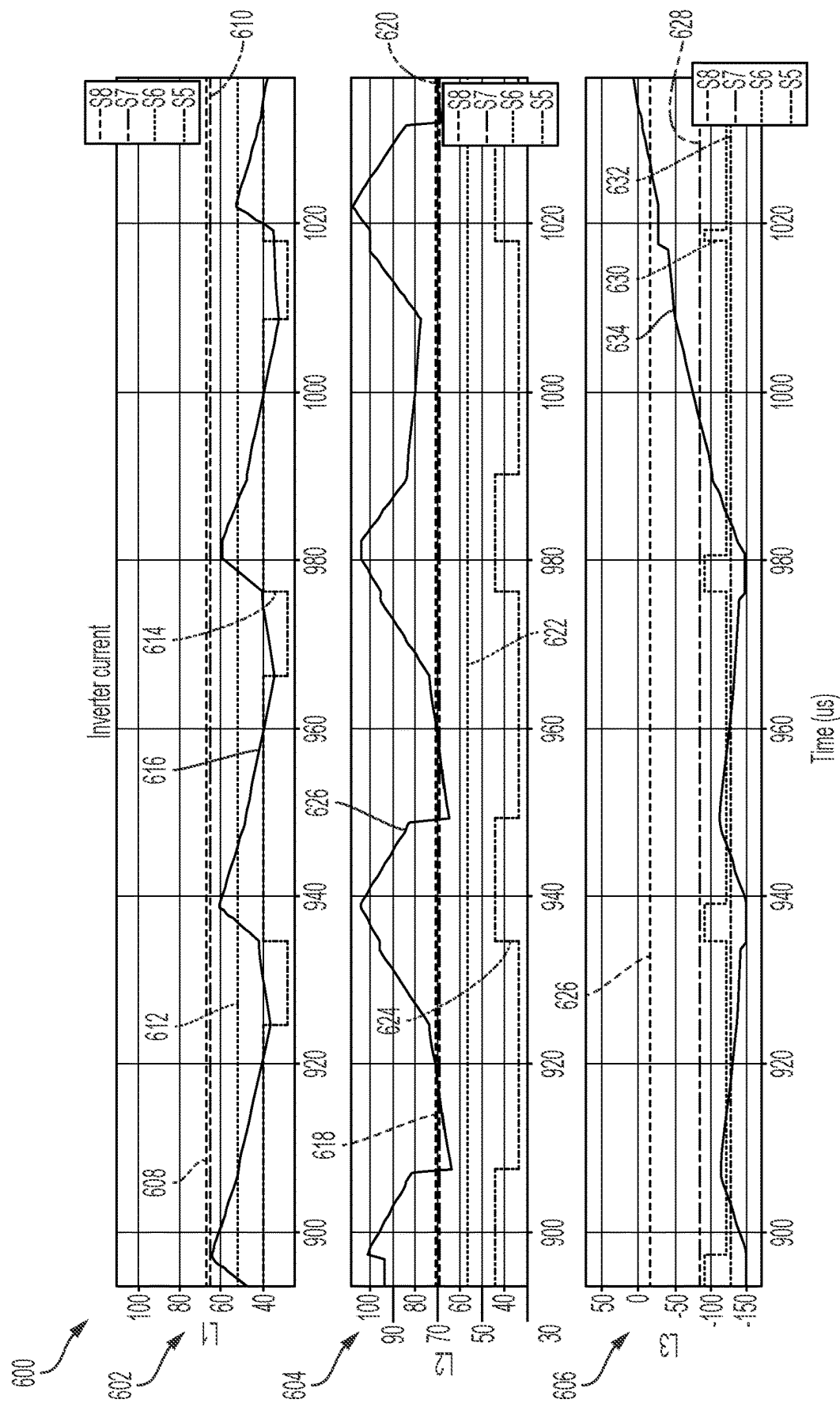


FIG. 6

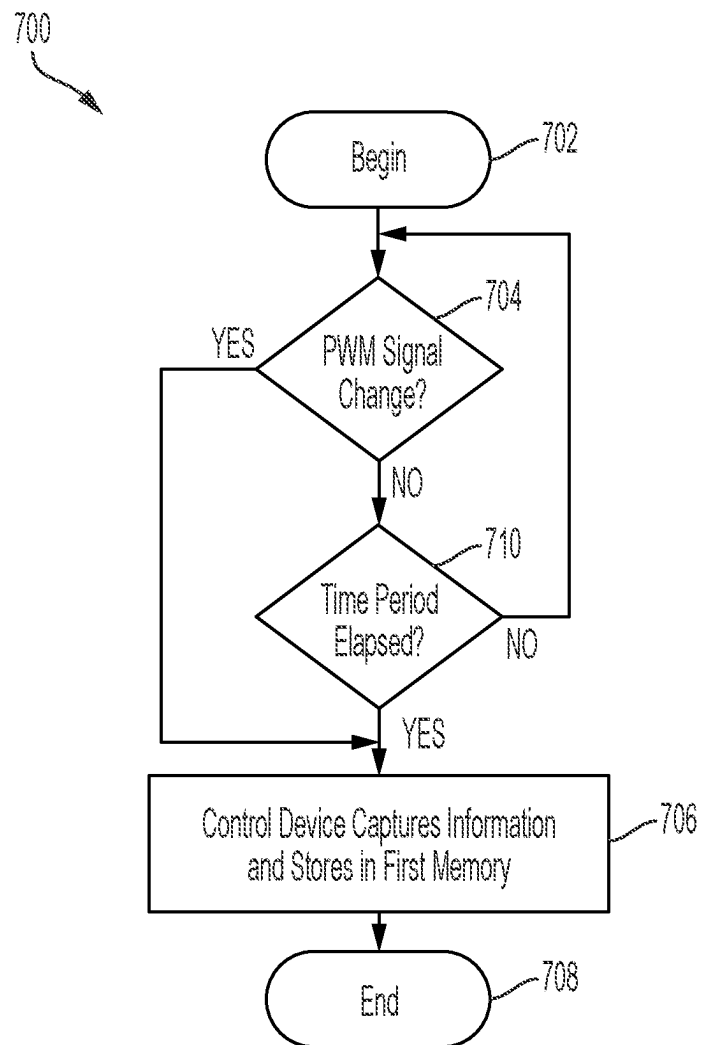


FIG. 7

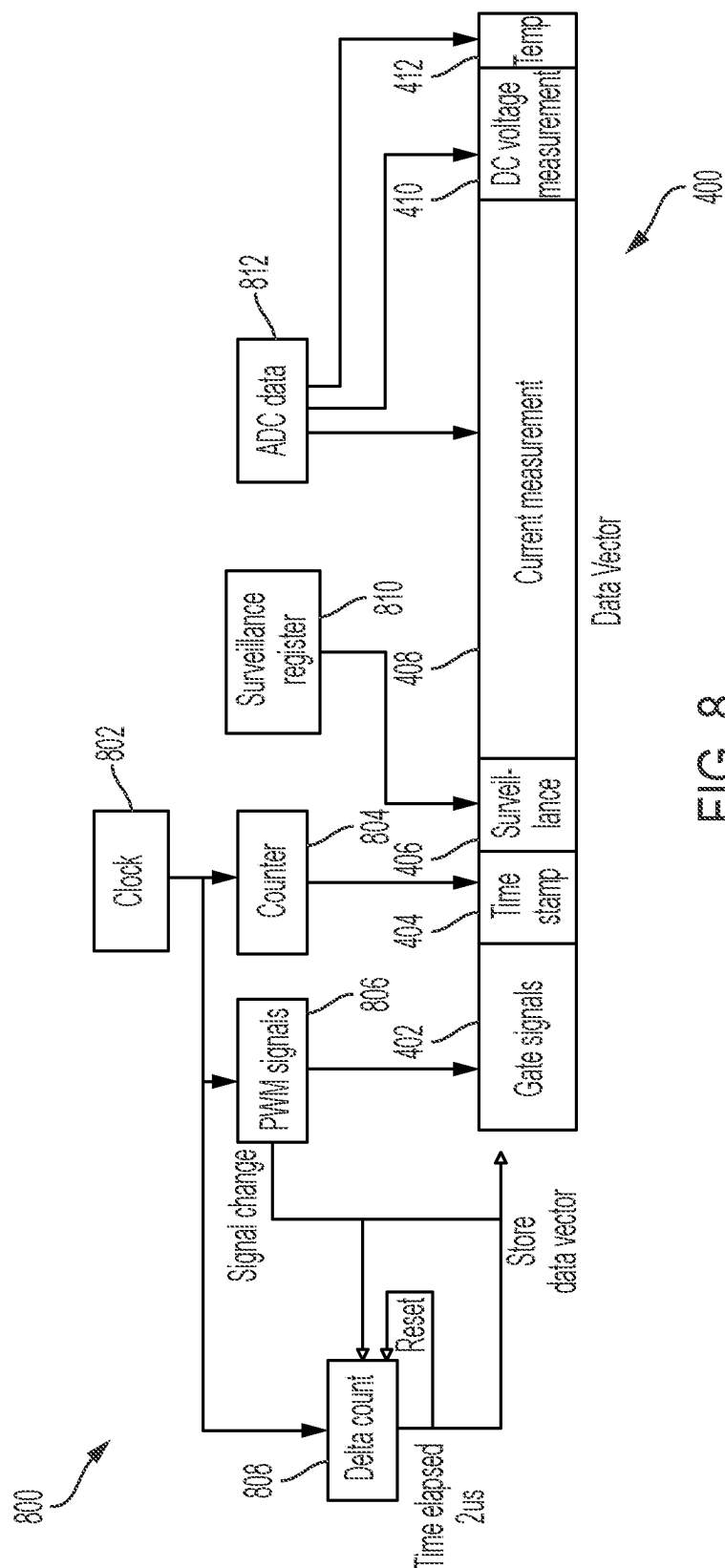


FIG. 8

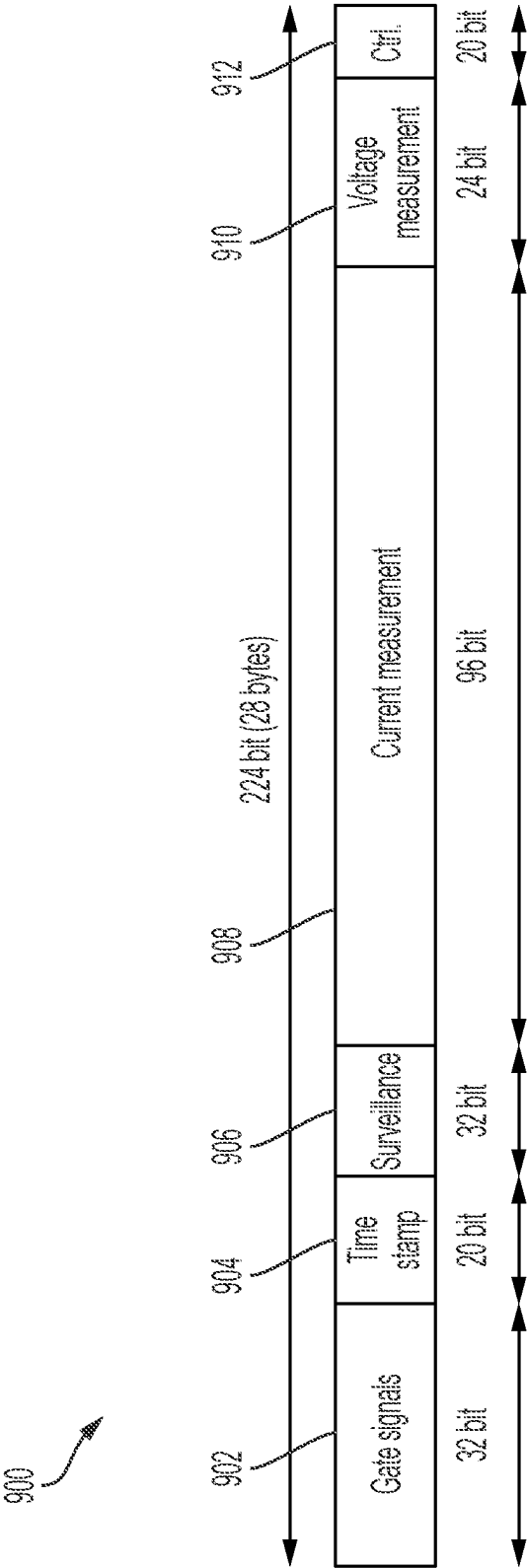


FIG. 9

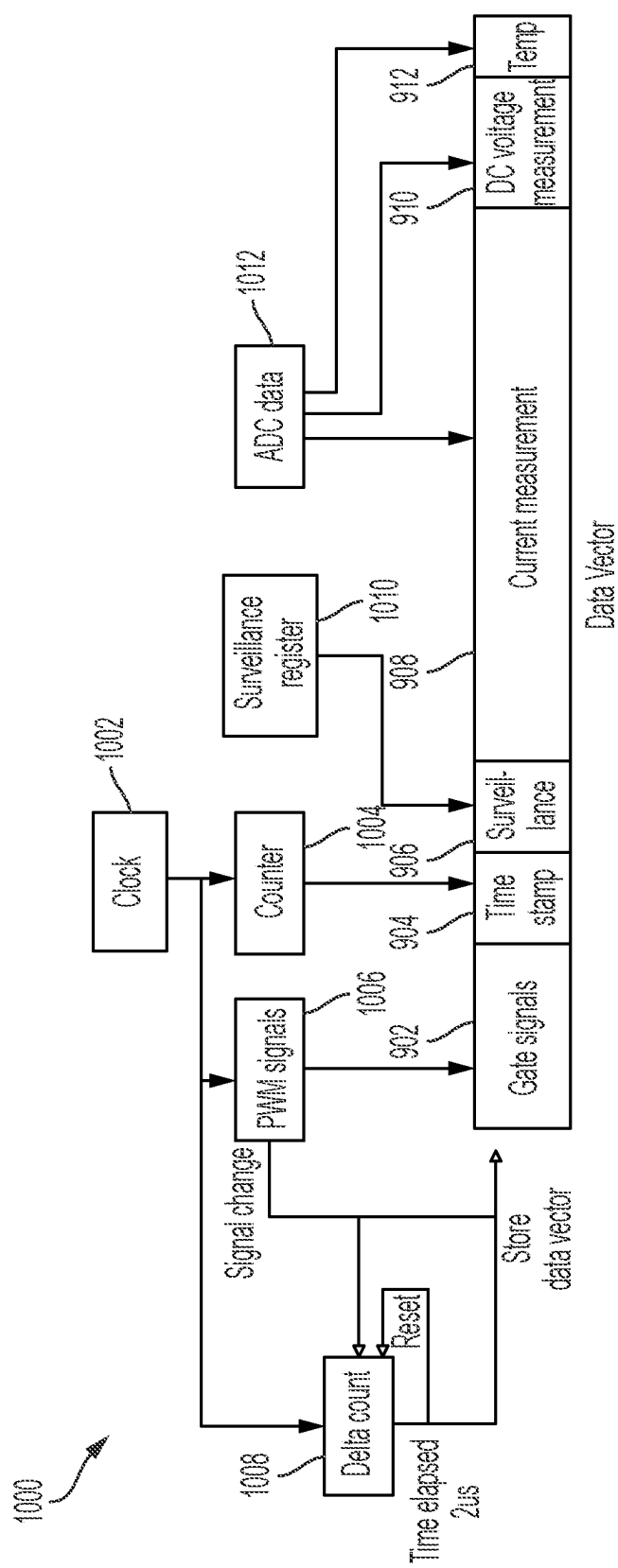


FIG. 10

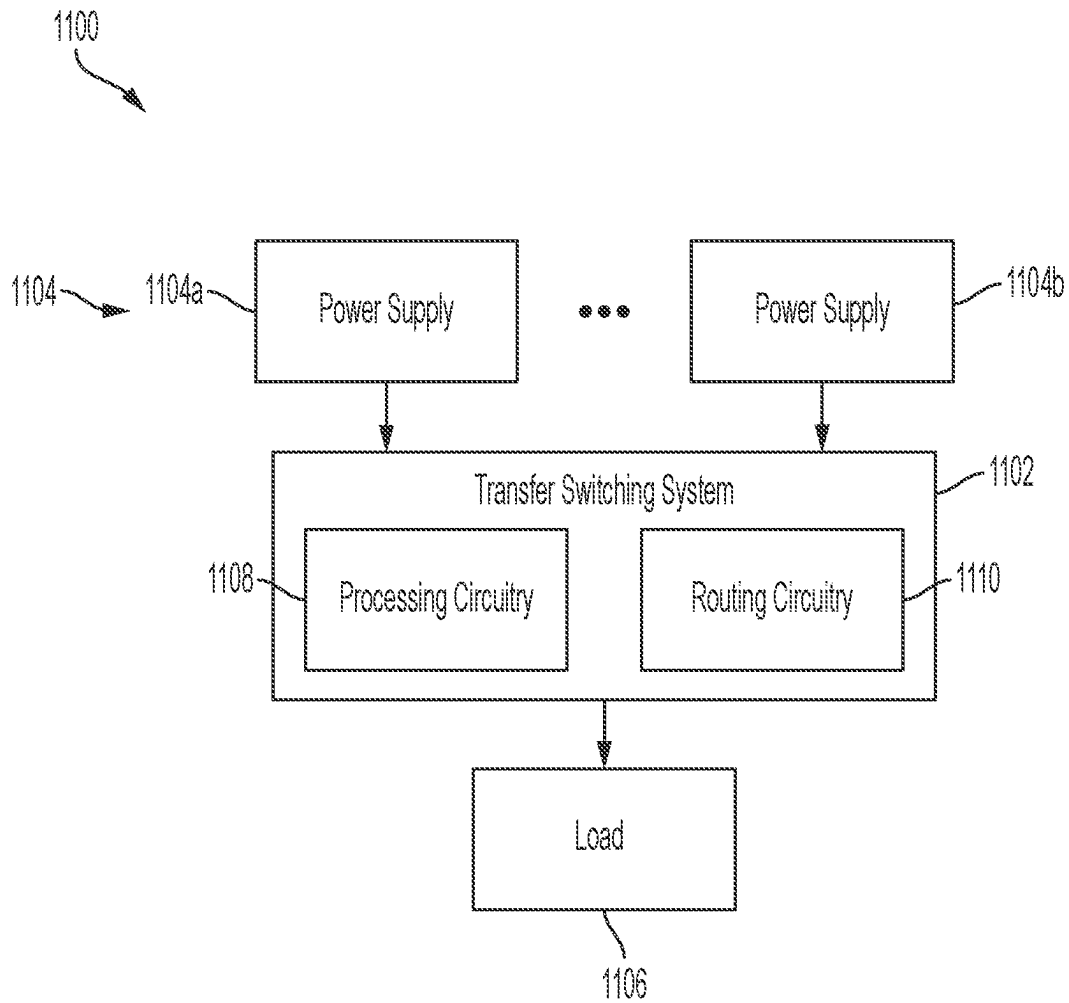


FIG. 11

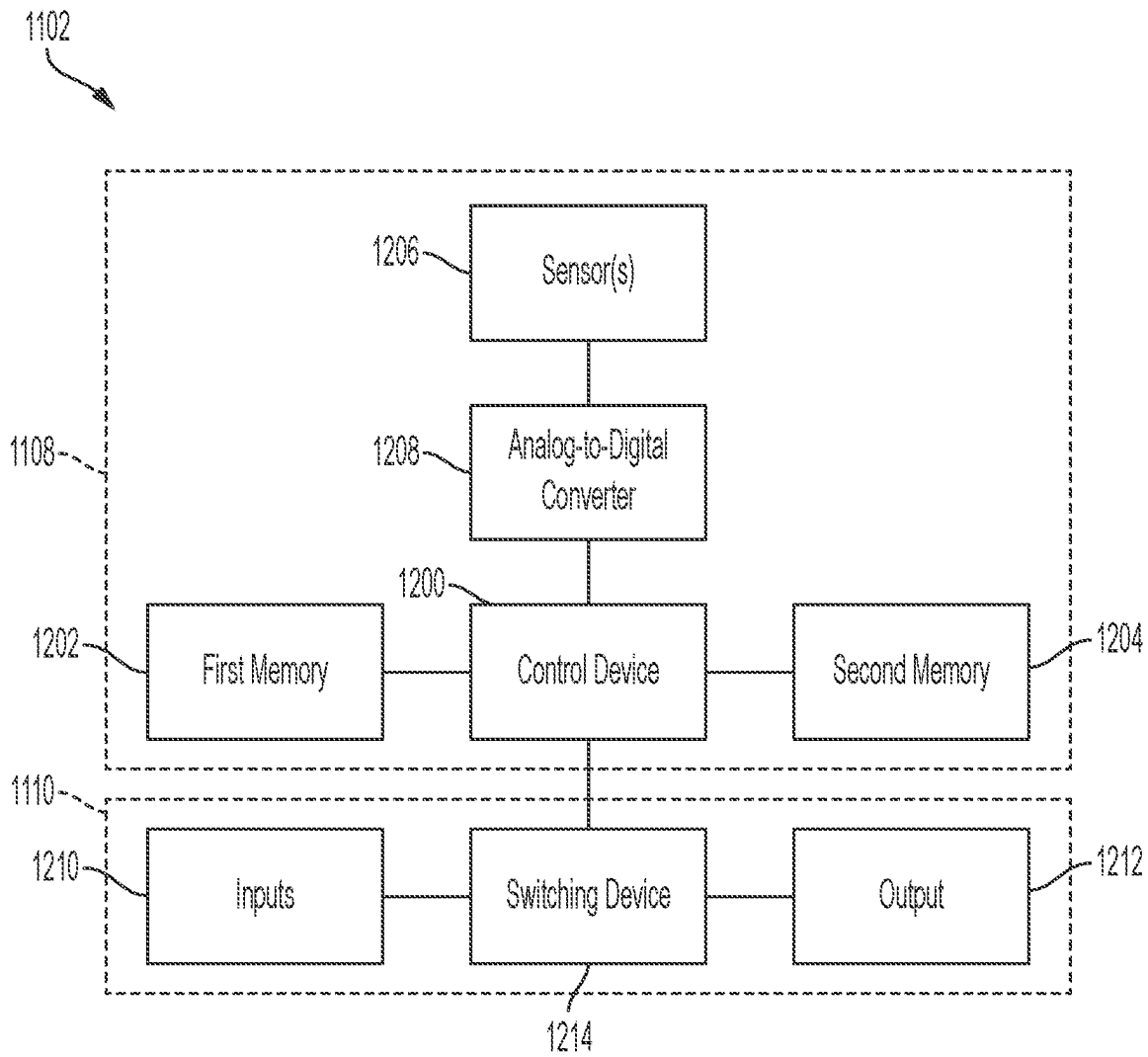


FIG. 12

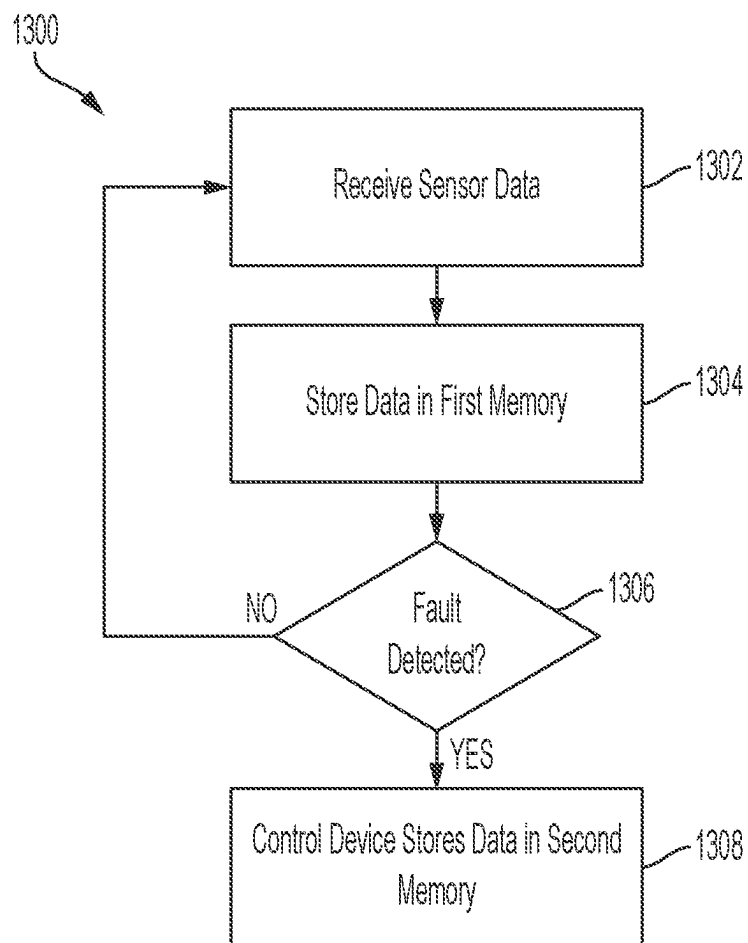


FIG. 13

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PWM CAPTURE FUNCTION FOR EVENT ANALYSIS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation-in-part of, and claims priority under 35 U.S.C. § 120 to, U.S. application Ser. No. 17/204,355, titled “PWM CAPTURE FUNCTION FOR EVENT ANALYSIS,” filed on May 5, 2022, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application Ser. No. 63/110,167, titled “PWM CAPTURE FUNCTION FOR EVENT ANALYSIS,” filed on Nov. 5, 2020, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Field of the Disclosure

At least one example in accordance with the present disclosure relates generally to analyzing electrical-parameter information.

2. Discussion of Related Art

Many electrical devices include electrical switching devices, or “switches.” Electrical power converters and/or power-routing devices, for example, may include several switches configured to open or close according to control signals received from a control device. Control signals may include pulse-width modulation (PWM) signals configured to control a state of the switches to which the PWM signals are provided. Operation of an electrical device, such as a power converter, may therefore be controlled based on PWM signals provided to switches in the electrical device. Some electrical devices may be controlled based on switching signals other than PWM signals in addition to, or in lieu of, PWM signals.

SUMMARY

According to at least one aspect, an electrical device comprises electrical-device components, a first memory configured to store data temporarily, a second memory, the second memory being non-volatile memory, and a control device configured to acquire data indicative of at least one parameter of at least one electrical-device component of the electrical-device components responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more of the electrical-device components, and store the acquired data in the second memory responsive to determining that the fault condition exists.

In various examples, the first memory is a buffer. In at least one example, the buffer is a first-in, first-out buffer. In various examples, the second memory includes ferroelectric random-access memory. In at least one example, the electrical-device components include one or more switching devices. In various examples, the fault condition is indicative of a fault of at least one of the one or more switching devices. In at least one example, the control device is further configured to provide pulse-width modulation (PWM) signals to the one or more switching devices. In various examples, the sample condition includes a transition in at least one PWM signal of the PWM signals.

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In at least one example, the sample condition includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the at least one electrical-device component was most recently acquired by the control device. In various examples, the electrical device comprises a microprocessor configured to provide the acquired data to a visualization system to analyze the fault condition. In at least one example, the electrical-device components include a peak-current detector, and wherein the fault condition includes a peak current being exceeded. In various examples, the electrical-device components include a desaturation detector, and wherein the fault condition includes a desaturation condition being detected.

In at least one example, the acquired data indicative of the at least one parameter of the at least one electrical-device component includes at least one of current data, voltage data, or temperature data. In various examples, the electrical-device components include a plurality of switching devices, and wherein the current data is indicative of a current through each switching device of the plurality of switching devices. In at least one example, the electrical-device components include a plurality of switching devices, and wherein the voltage data is indicative of a voltage across each switching device of the plurality of switching devices. In various examples, the temperature data is indicative of an ambient temperature of the at least one electrical-device component of the electrical-device components.

According to at least one aspect, a non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for operating an electrical device having electrical-device components, a first memory configured to store data temporarily, and a second memory is provided, the second memory being non-volatile memory, and the sequences of computer-executable instructions including instructions that instruct at least one device to acquire data indicative of at least one parameter of at least one electrical-device component of the electrical-device components responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more of the electrical-device components, and store the acquired data in the second memory responsive to determining that the fault condition exists.

In various examples, the sequences of computer-executable instructions include instructions that instruct at least one processor to provide pulse-width modulation (PWM) signals to one or more switching devices, and wherein the sample condition being met includes a transition in at least one PWM signal of the PWM signals. In at least one example, the sample condition being met includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the at least one electrical-device component was most recently acquired.

According to at least one example, a method of assembling an electrical device is provided comprising providing electrical-device components, a first memory configured to store data temporarily, and a second memory, the second memory being non-volatile memory, providing a control device configured to acquire data indicative of at least one parameter of at least one electrical-device component of the electrical-device components responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more of the electrical-device components, and store the acquired data in the second memory responsive to deter-

mining that the fault condition exists, and coupling the control device to the electrical-device components, the first memory, and the second memory.

According to at least one aspect of the disclosure, an electrical device is provided comprising an input configured to receive power from at least one power supply, an output configured to provide the power to at least one load, a first memory configured to store data temporarily, a second memory, the second memory being non-volatile memory, and at least one control device configured to acquire data indicative of at least one parameter of the power responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components, and store the acquired data in the second memory responsive to determining that the fault condition exists.

In at least one example, the first memory is a buffer. In at least one example, the buffer is a first-in, first-out buffer. In at least one example, the second memory includes ferro-electric random-access memory. In at least one example, the sample condition includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the power was most recently acquired by the at least one control device. In at least one example, the electrical device includes a microprocessor configured to provide the acquired data to a visualization system to analyze the fault condition. In at least one example, the fault condition includes a peak current being exceeded. In at least one example, the fault condition includes a desaturation condition being detected.

In at least one example, the acquired data indicative of the power includes at least one of current data, voltage data, or temperature data. In at least one example, the acquired data includes the current data, and wherein the current data is indicative of an input current received at the input. In at least one example, the acquired data includes the voltage data, and wherein the voltage data is indicative of an input voltage received at the input. In at least one example, the fault condition is indicative of a fault upstream of the input. In at least one example, determining that the fault condition exists includes determining that a frequency of voltage at the input is outside of a threshold range.

According to at least one aspect of the disclosure, a non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for operating an electrical device having an input, an output, a first memory configured to store data temporarily, and a second memory, the second memory being non-volatile memory is provided, and the sequences of computer-executable instructions including instructions that instruct at least one device to acquire data indicative of at least one parameter of the power responsive to a sample condition being met, store the acquired data in the first memory, determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components, and store the acquired data in the second memory responsive to determining that the fault condition exists.

In at least one example, the sample condition includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the power was most recently acquired. In at least one example, determining that the fault condition exists includes determining that a frequency of voltage at the input is outside of a threshold range. In at least one example, determining that the fault condition exists includes determining that a peak current is

being exceeded. In at least one example, determining that the peak current is being exceeded includes determining that a current received at the input exceeds a current threshold. In at least one example, determining that the fault condition exists includes determining that a voltage is outside of a threshold range. In at least one example, determining that the voltage is outside of the threshold range includes determining that an input voltage at the input is outside of the threshold range.

According to at least one aspect of the disclosure, a method for operating an electrical device having an input, an output, a first memory configured to store data temporarily, and a second memory, the second memory being non-volatile memory is provided, the method comprising acquiring data indicative of at least one parameter of the power responsive to a sample condition being met, storing the acquired data in the first memory, determining that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components, and storing the acquired data in the second memory responsive to determining that the fault condition exists.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

FIG. 1 illustrates a block diagram of an electrical device according to an example;

FIG. 2 illustrates a block diagram of components of a type of power module, such as an uninterruptible power supply, which may be included within the electrical device of FIG. 1 according to an example;

FIG. 3 illustrates a process of acquiring electrical device information to enable analysis of a cause of a fault condition of one or more switching devices according to an example;

FIG. 4 illustrates a schematic diagram of a data vector according to an example;

FIG. 5 illustrates a block diagram of components of a visualization system according to an example;

FIG. 6 illustrates a visualization of information indicative of the electrical device of FIG. 1 according to an example;

FIG. 7 illustrates a process of capturing information based on a sample condition;

FIG. 8 illustrates a schematic view of a data-vector-generation scheme according to an example;

FIG. 9 illustrates a schematic diagram of a data vector according to an example;

FIG. 10 illustrates a schematic view of a data-vector-generation scheme according to an example;

FIG. 11 illustrates a block diagram of a power system according to an example;

FIG. 12 illustrates a block diagram of processing circuitry according to an example; and

FIG. 13 illustrates a process of operating processing circuitry according to an example.

DETAILED DESCRIPTION

Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

As discussed above, certain electrical devices, such as power converters, may include one or more electrical switches configured to open and close to prevent and allow, respectively, current to pass therethrough. A control unit may control a state of each switch by providing PWM signals to each switch. Over time, one or more switches may experience a fault condition, such as by being unable to open or close in response to control signals from the control unit. Operation of an electrical device including a faulty switch may be negatively impacted by the fault condition.

In various examples, switches in a device are monitored for a fault condition. In various examples, a cause of the fault condition may be determined. Determining a cause of the fault condition may be facilitated by analyzing certain device information descriptive of the device in which the switches are implemented, such as current information, voltage information, temperature information, PWM signal information, timestamp information, and/or surveillance information indicative of a detected error condition, if any. Samples of such device information may be analyzed for a period of time before, during, and/or after the fault occurs to diagnose a cause of the fault. In some cases, device information is repeatedly sampled and stored such that, if a fault condition is detected, the device information acquired before, during, and/or after the fault condition may be analyzed to diagnose a cause of the fault. This may be easier or more practical than predicting a fault condition in

advance of the fault condition occurring. However, repeatedly acquiring and storing device information may require significant storage resources.

Examples provided herein enable device information to be sampled and temporarily stored responsive to a sample condition being met. If a fault condition is thereafter detected, then the temporarily stored information may be stored on a non-temporary basis to perform root-cause analysis. For example, the information may be stored in non-volatile memory. In some examples, there are two sample conditions that prompt sampling and storage of the device information. In other examples there may be more or fewer than two sample conditions. These sample conditions may be provided in a device in which electrical switches are present. The electrical switches may be controlled based on PWM signals.

A first sample condition includes a PWM-state transition, such as from a logic HIGH value to a logic LOW value or a logic LOW value to a logic HIGH value. A second sample condition includes a threshold period of time having elapsed since device information was last acquired. In this example, therefore, device information is acquired responsive to a PWM state switching or after a threshold period of time if no PWM state switch has occurred during the threshold period of time.

The device information may initially be stored temporarily responsive to the sample condition being met. The temporarily stored device information may be transferred to non-volatile storage if a triggering event occurs, such as a fault condition being detected. For example, the device information may be temporarily stored in a first-in, first-out (FIFO) buffer such that, if the buffer is full when new device information is acquired, the oldest device information in the buffer is overwritten. A size of the buffer may be selected to store a desired amount of information to determine a cause of the fault condition. The buffer may be considered to store data “temporarily” inasmuch as stored data is overwritten shortly after being stored, unless the data is moved to other, non-volatile memory or storage for indefinite storage (for example, for storage until explicitly selected by a user for deletion). Accordingly, examples disclosed herein enable a relatively small amount of information to be repeatedly temporarily acquired and stored until a triggering event is met, responsive to which the temporarily stored information is non-temporarily stored in non-volatile memory to facilitate root-cause analysis. The amount of data stored at any one time may therefore be minimized.

Current fault-detection and/or -analysis systems may not store certain information that may be beneficial in analyzing fault conditions, or may store significant volumes of such information. Such systems may operate inefficiently, because less information may be collected than may be desirable, or so much information may be collected that memory and/or storage requirements may be significant. This is a technical problem. An exemplary embodiment of a fault-detection and/or analysis system as described herein may acquire data indicative of at least one parameter of one or more electrical-device components responsive to a sample condition being met, store the acquired data in a first memory configured to store data temporarily, and store the acquired data in non-volatile storage responsive to determining that a fault condition exists. The fault condition may be indicative of a fault of one or more of the electrical-device components. At least this foregoing combination of features comprises a fault-detection and -analysis system that serves as a technical solution to the foregoing technical problem by acquiring and storing useful information without

requiring significant memory and/or storage resources. This technical solution is not routine and is unconventional. This technical solution is a practical application of the fault-detection and -analysis system design that solves the foregoing technical problem and constitutes an improvement in the technical field of electrical device systems at least by enabling faults to be analyzed effectively without requiring significant resources, such as memory and/or storage resources.

FIG. 1 illustrates a block diagram of an electrical device 100 according to an example. The electrical device 100 may include any device that operates using electrical power. For example, the electrical device 100 may be a power device, such as a power converter. In one example, a power converter may be configured to convert electrical power in a three-phase system. Each phase may have ten switching devices, such that the power converter includes 30 total switching devices. In other examples, the electrical device 100 may be a device other than a power device, and/or may include more or fewer than 30 switches.

The electrical device includes an analog-to-digital converter (ADC) 102, a control device 104 including a buffer 106, a microprocessor 108, a memory 110, which may include non-volatile memory technology such as ferroelectric random-access memory (FRAM) technology or other non-volatile memory technology, a first set of one or more sensors 112 (including, for example, temperature sensors, current sensors, voltage sensors, and so forth), and additional components 114 including one or more switching devices 116, one or more surveillance circuits 118, and a second set one or more sensors 120 (including, for example, temperature sensors, current sensors, voltage sensors, and so forth). The one or more surveillance circuits 118 include one or more circuits configured to monitor the additional components 114 for certain conditions or events, such as a peak current detector 120 configured to detect a peak current of one or more of the switching devices 116 being exceeded, a desaturation detector 122 configured to detect a desaturation condition of one or more of the switching devices 116 being experienced, and so forth. The one or more surveillance circuits 118 may send surveillance information indicative of any detected conditions or events (such as a fault condition), if any, to the control device 104. In some examples, the control device 104 may be, for example, a field-programmable gate array (FPGA), or another device configured to control a device, such as a microcontroller, application-specific integrated circuit, and so forth.

The additional components 114 may include components specific to an application of the electrical device 100. For example, where the electrical device 100 is a power module configured to provide and store electrical power, the additional components 114 may include components configured to enable or facilitate the providing and storing electrical power, including the switching devices 116. The switching devices 116 may include, for example, metal-oxide semiconductor field-effect transistors (MOSFETs), bipolar junction transistors (BJTs), insulated-gate bipolar transistors (IGBTs), or other types of switching devices, which may depend on a particular implementation of the additional components 114.

FIG. 2 illustrates a block diagram of components of a type of power module 200, such as an uninterruptible power supply (UPS), which may be included within the additional components 114 of FIG. 1 according to an example. FIG. 2 may illustrate only a subset of the components of the power module 200. For example, the power module 200 may be a three-phase power module, and FIG. 2 may only illustrate

components of a single phase of the three phases for the sake of clarity. Other phases of the power module may be similarly configured. Accordingly, FIG. 2 may illustrate only a subset of the additional components 114 of FIG. 1.

The power module 200 includes an input 202, a rectifier 204, a direct-current (DC) bus 206 including at least one control switch 208 (“control switch 208”), a DC/DC converter 210, a battery 212, an inverter 214, and an output 216. The input 202 is configured to receive input power, such as alternating-current (AC) power from a utility mains source. The rectifier 204 is configured to rectify the input power, which may include providing power-factor correction (PFC) to the input power. The DC bus 206 is configured to conduct DC current between components connected thereto, including the rectifier 204, the inverter 214, and the DC/DC converter 210, and the control switch 208 is configured to control how the DC current passes through the DC bus 206. For example, the control switch 208 may have a first position in which DC current is provided from the rectifier 204 to the DC/DC converter 210 and/or the inverter 214, and a second position in which DC current is provided from the DC/DC converter 210 to the inverter 214. The control switch 208 may be in the first position when AC power received at the input 202 is acceptable (for example, having a voltage and/or current level within an acceptable range) and may be in the second position when AC power received at the input 202 is not acceptable (for example, having a voltage and/or current level not within an acceptable range).

The DC/DC converter 210 is configured to convert DC power from a first voltage and/or current to a second voltage and/or current. The DC/DC converter 210 may be bidirectional, such that power received from the DC bus 206 is converted and provided to the battery 212 in a first direction, and such that power received from the battery 212 is converted and provided to the DC bus 206 in a second direction. The DC/DC converter 210 may provide power to the battery 212 when the AC power received at the input 202 is acceptable, and may provide power from the battery 212 when the AC power received at the input 202 is not acceptable. The battery 212 is configured to receive electrical power from the DC/DC converter 210, store the electrical power, and discharge stored electrical power to the DC/DC converter 210. The inverter 214 is configured to receive DC power from the DC bus 206, invert the DC power to AC power, and provide the AC power to the output 216. In some examples, the battery 212 is external to, and electrically coupled to, the power module 200. For example, the power module 200 may include a power interface coupled to, or part of, the DC/DC converter 210 to enable power to be exchanged between the power module 200 and the battery 212. The output 216 provides the AC power to a device that may be connected thereto, such as a load.

As discussed above, the additional components 114 may include the switching devices 116. For example, the rectifier 204, the DC bus 206, the DC/DC converter 210, and the inverter 214 may each include one or more switching devices. More particularly, the rectifier 204 may include one or more rectifier switches 218, the DC bus 206 may include the control switch 208, the DC/DC converter 210 may include one or more converter switches 220, and the inverter 214 may include one or more inverter switches 222. The switching devices 116 may include some or all of the switches 208, 218-222.

The control device 104 may generate PWM signals and provide the PWM signals to the switching devices 116 to control the switching devices 116. One or more of the switching devices 116 may at some time experience a fault

condition, which may include any condition, such as an unintended condition, that affects operation of the switching devices **116**. For example, the fault condition may include a condition that negatively affects operation of the switching devices **116**. Such fault conditions may include, for example, a temperature of a switching device being exceeded, or other fault conditions.

As discussed above, in various examples the fault conditions may be identified and a cause of the fault condition may be determined based on information descriptive of operation of the electrical device **100** before, during, and/or after the fault condition occurs. Electrical device information may be repeatedly acquired over time and, if a fault condition occurs, electrical device information descriptive of the electrical device **100** close in time to the fault condition may be extracted. The extracted information may be subsequently analyzed to determine, for example, a cause of the fault condition.

FIG. **3** illustrates a process **300** of acquiring electrical device information to enable analysis of a cause of a fault condition of one or more switching devices according to an example. The process may be executed in connection with the electrical device **100** and components thereof, such as the switching devices **116**, which may include the switches **208**, **218-222**.

At act **302**, the process **300** begins.

At act **304**, the control device **104** acquires data and stores the data in a first memory responsive to a sample condition being met. An example of act **304**, including examples of sample conditions, is discussed in greater detail below with respect to FIG. **7**. The first memory may be configured to store data temporarily. For example, the first memory may be the buffer **106**, which may be a FIFO buffer in some examples. In other examples, the first memory may be a different type of memory or storage, such as a last-in, first-out (LIFO) buffer, a different type of temporary memory other than a buffer, a memory configured to store data non-temporarily, and so forth.

A portion of the information captured at act **304** may be acquired by the control device **104** itself and/or other components of the electrical device **100**, such as the ADC **102**. For example, the ADC **102** may repeatedly (for example, periodically, continuously, aperiodically, and so forth) sample one or more sensors (including, for example, the first set of one or more sensors **112**, the second set of one or more sensors **120**, or both) for one or more parameters of the switching devices **116** and/or the electrical device **100** as a whole. For example, where the electrical device **100** is a power module, the ADC **102** may sample a current of each phase of the output (including, for example, the output **216**), a current of each phase of the input (including, for example, the input **202**), a current at each pole of the battery **212**, a voltage on a respective DC bus (including, for example, the DC bus **206**) of each rail between each respective rectifier (including, for example, the rectifier **204**) and inverter (including, for example, the inverter **212**), a voltage at an input (including, for example, an AC input voltage at the input **202**), a voltage at an output (including, for example, an AC output voltage at the output **216**), a voltage at the battery **212**, and so forth.

In some examples, the ADC **102** may sample one or more sensors to acquire each of the foregoing examples of information at act **304**. In other examples, the ADC **102** may sample a subset of the one or more sensors and/or acquire a subset of the foregoing examples of information at act **304**. For example, the ADC **102** may capture each of the examples of current information identified above at each

execution of act **304**, but may alternate between capturing different subsets of the examples of voltage information. In executing act **304** at a first time in one example, the voltage information that the ADC **102** may capture may include, for example, the voltage on the respective DC bus of each rail between each respective rectifier and inverter. In executing act **304** at a second time in one example, the voltage information that the ADC **102** may capture may include, for example, the voltage at the input. In executing act **304** at a third time in one example, the voltage information that the ADC **102** may capture may include, for example, the voltage at the output. In executing act **304** at a fourth time in one example, the voltage information that the ADC **102** may capture may include, for example, the voltage at the battery **212**. In executing act **304** at a fifth time in one example, the voltage information that the ADC **102** may capture may include, for example, the voltage on the respective DC bus of each rail between each respective rectifier and inverter, as was performed at the first time. This cycle may then repeat, such that varied voltage information may be captured over time without capturing all of the voltage information at any one time. In some examples, act **304** may be repeatedly executed at a frequency that is high relative to the frequency of voltage being measured (for example, AC voltage received at the input **202**). It is to be appreciated that the examples provided above are non-limiting, and that in other examples, different combinations of information may be acquired at act **304** in different orders and/or frequencies. The ADC **102** may also acquire information such as a temperature of the electrical device **100** and/or the additional components **114** from a sensor, such as a temperature sensor, of either or both of the first and second sets of sensors **112**, **120**. The information acquired by the ADC **102** may be converted to a digital format and provided to the control device **104**. Capturing the information at act **304** may include the control device **104** controlling the ADC **102** to acquire and provide the information. In some examples, capturing the information at act **304** may include the control device **104** requesting that the ADC **102** provide the information to the control device **104**. As discussed in greater detail below, the control device **104** may request the ADC information or control the ADC **102** to provide the information responsive to a period of time elapsing or a PWM-signal state change.

In storing the electrical device information at act **304**, the control device **104** generates a data vector. FIG. **4** illustrates an example of a data vector **400** generated by the control device **104** according to an example. The data vector **400** includes gate signal information **402**, timestamp information **404**, surveillance information **406**, current information **408**, DC voltage measurement information **410**, and temperature information **412**, collectively referred to as "device information." In one example, the data vector **400** is 216 bits long with the gate signal information **402** being represented by 32 bits, the timestamp information **404** being represented by 20 bits, the surveillance information **406** being represented by 32 bits, the current measurement information **408** being represented by 96 bits, the DC voltage measurement information **410** being represented by 24 bits, and the temperature information **412** being represented by 12 bits, although in other examples, the data vector **400** and portions thereof may be represented by a different number of bits and may include additional, less, or different information.

The current information **408**, DC voltage measurement information **410**, and temperature information **412** may be included in the information received from the ADC **102**. The ADC **102**, in turn, may acquire the current information **408**,

DC voltage measurement information **410**, and temperature information **412** by sampling one or more sensors, such as the first set of one or more sensors **112**, the second set of one or more sensors **120**, or both. Other information in the data vector **400** may be generated or acquired by the control device **104** itself. For example, the gate signal information **402** includes information indicative of PWM signals provided by the control device **104** to the switching devices **116**. The gate signal information **402** may indicate whether a PWM signal provided to each corresponding switching device was at a logical HIGH or logical LOW value when the data vector **400** was generated. The timestamp information **404** is indicative of a time at which the data vector **400** was generated. For example, the timestamp information **404** may include a clock value, a number of clock cycles which may be used to determine a clock value, or any other information that may be used to determine a time at which the data vector **400** was generated.

The surveillance information **406** indicates a fault condition, if any, detected by the one or more surveillance circuits **118** and/or the control device **104** based on information received from the one or more surveillance circuits **118**, such as the peak current detector **122**, the desaturation detector **124**, and so forth. The control device **104** may receive the information from the ADC **102** and analyze the information to determine if a fault condition exists. For example, the control device **104** may determine that a fault condition exists based on a voltage or current value being outside of an acceptable range of values. In other examples, the control device **104** may not determine that a fault condition exists, or may not analyze the ADC information for a fault condition at all, and may instead receive the surveillance information **406** indicating whether a fault condition exists directly from the one or more surveillance circuits **118**.

The control device **104** stores the data vector **400** in the first memory, which may be the buffer **106** in an example. In some examples, the buffer **106** may be configured to store multiple data vectors. The buffer **106** may be a FIFO buffer, such that if the FIFO buffer is full when the data vector **400** is stored in the FIFO buffer, an oldest data vector stored in the FIFO buffer is overwritten. A size of the FIFO buffer may be selected based on a desired number of data vectors to be stored.

FIG. 9 illustrates another example of a data vector **900** according to another example. The data vector **900** is similar to the data vector **400** and includes gate signal information **902**, timestamp information **904**, surveillance information **906**, and current information **908**, which may be substantially similar to the gate signal information **402**, the timestamp information **404**, the surveillance information **406**, and the current information **408**, respectively. The data vector **900** further includes voltage measurement information **910** and control information **912**. The information **902-912** is collectively referred to as "device information." In one example, the data vector **900** is 224 bits long with the gate signal information **902** being represented by 32 bits, the timestamp information **904** being represented by 20 bits, the surveillance information **906** being represented by 32 bits, the current measurement information **908** being represented by 96 bits, the voltage measurement information **910** being represented by 24 bits, and the control information **912** being represented by 20 bits, although in other examples, the data vector **900** and portions thereof may be represented by a different number of bits and may include additional, less, or different information.

The information included in the voltage measurement information **910** may differ from the information included in

the DC voltage measurement information **410**. As discussed above, in some examples of act **304**, the ADC **102** may capture different information in successive executions of act **304**. For example, the ADC **102** may capture different voltage information in successive executions of act **304**, such as by capturing information indicative of a DC voltage at the DC bus **206** at a first time, information indicative of an AC voltage at the input **202** at a second time, and so forth. In some examples, the ADC **102** may include, or be coupled to, a multiplexer having a plurality of inputs, each configured to receive different voltage information, such as the examples of voltage information provided above. The multiplexer may further include an output, and the multiplexer may route an input of the plurality of inputs to the output, where the output of the multiplexer may provide the voltage information **910**. Accordingly, in some examples, the voltage information **910** includes voltage information that varies over time as act **304** is repeatedly executed and data vectors, such as the data vector **900**, are repeatedly generated.

The information included in the control information **912** may differ from the information included in the temperature information **412**. In some examples, the control information **912** may include temperature information substantially identical to what is included in the temperature information **412**. The control information **912** may further include additional information. For example, where the voltage measurement information **910** includes various different types of voltage information that change over time, the control information **912** may further indicate what type of voltage information is indicated by the voltage measurement information **910**, such as by indicating that the voltage measurement information **910** is indicative of an AC voltage at the input **202**, an AC voltage at the output **216**, and so forth. In some examples, the control information **912** may further include other information, such as information to optimize other components of the electrical device **100**, including the buffer **106**.

Returning to FIG. 3, after the control device **104** stores the data vector at act **304**, the process **300** continues to act **306**. For purposes of explanation only, the following examples may refer to aspects of the data vector **400**, that is, the following examples may be provided where the data vector **400** is stored at act **304**. It is to be appreciated that, in other examples, the following principles are also applicable to other types of data vectors, such as the data vector **900**.

At act **306**, the control device **104** determines whether a fault condition has been detected based on information in the data vector **400** (including, for example, the surveillance information **406** received from the one or more surveillance circuits **118**). If a fault condition is determined not to have occurred (**306 NO**), then the process **300** returns to act **304**. The process **300** continues until a fault condition (**306 YES**) is detected, at which point the process **300** continues to act **308**. As discussed in greater detail below, however, the process **300** may be repeatedly executed even if a fault condition is detected (**306 YES**).

At act **308**, the control device **104** stores the contents of the first memory in a second memory and/or storage. For example, the control device **104** may provide the contents of the buffer **106** to the microprocessor **108**, and the microprocessor **108** may store the contents of the buffer **106** in the second memory. The control device **104** may provide the contents of the buffer **106** to the microprocessor **108** immediately upon determining that a fault condition exists (**306 YES**) or after a delay, such as after one or more sample periods. The second memory may be non-volatile memory. For example, the second memory may be or include the memory **110**, which may include non-volatile memory, such

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as FRAM memory or another type of non-volatile memory. The control device **104** may provide the contents of the buffer **106** (for example, one or more data vectors, which may include the data vector **400** and/or the data vector **900**) to the microprocessor **108** in the form that the data vector(s) were initially stored in the buffer **106**, which may be referred to as a “raw” form. In other examples, the contents of the buffer **106** may be provided to the microprocessor **108** in another form, such as a converted form (for example, converted by the control device **104** prior to, during, or after storage in the buffer **106**). The microprocessor **108** may store the data in the memory **110** in the raw form, or another form.

At act **310**, the microprocessor **108** determines if the microprocessor **108** is operational and/or is likely to be non-operational shortly thereafter. Certain fault conditions may impact performance of the microprocessor **108**. For example, a fault condition may include a failure of a switching device that controls power input at an input of the electrical device **100** as a whole, or a component thereof, such as a power supply of the microprocessor **108**. Without the input power to power the microprocessor **108**, the microprocessor **108** may lack electrical power to remain operational for much longer, if at all, after the fault condition occurs. If the microprocessor **108** is not operational, or is likely to not be operational for much longer because of the fault condition (**310 NO**), then the determination is repeatedly executed until it is determined that the microprocessor is operational and is unlikely to be non-operational shortly thereafter (**310 YES**). For example, if the fault condition does not affect the ability of the microprocessor **108** to remain operational, or if the fault condition has ceased occurring and is no longer adversely impacting operation of the microprocessor **108**, then the microprocessor **108** may be determined to be operational (**310 YES**). The process **300** then continues to act **312**.

At act **312**, the microprocessor **108** converts the raw information stored in the memory **110** to an alternate format and stores the converted information in the same or a different memory. For example, the microprocessor **108** may convert the raw data to another format, such as a JavaScript Object Notation (JSON) file format, and store the converted data in another memory, such as a static random-access memory (SRAM) memory. Although the microprocessor **108** may convert the raw data to a format other than a JSON file format, such as a comma-separated value (CSV) file format or other file format, examples are provided with respect to a JSON file format for purposes of explanation.

In some examples, the microprocessor **108** may include additional information, along with the raw data, in the JSON file, such as serial-number information, conversion information, and/or other information. In some examples, the memory **110** may include the SRAM memory and the FRAM memory discussed above. In other examples, the memory **110** may include the SRAM memory or the FRAM memory, and the electrical device **100** may include additional memories, which may be SRAM memory, FRAM memory, or other types of memory, such as flash memory. Converting the data may include the microprocessor **108** applying a scale value for time, current, voltage, and temperature, segmenting the data for different devices to which the microprocessor **108** is connected, and exporting the data into the JSON file for storage.

At optional act **314**, the microprocessor **108** may provide the converted data, which may be included in a JSON file, to another storage, which may be external storage in a device other than the electrical device **100**. The converted data may

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be provided via a wired or wireless communication connection to the external storage. The external storage may be easily accessible for entities, such as human operators, to review the information included in the converted data to analyze a cause of the fault condition that triggered the control device **104** providing the raw data to the microprocessor **108**. The electrical device **100** may provide the converted data to the external storage responsive to a device including the external storage requesting the converted data, and/or other information stored by the electrical device **100**, from the electrical device **100**. In another example, the electrical device **100** may send such information to the device including the external storage periodically, aperiodically, or continuously, either directly or via one or more intermediate devices, with or without a prompt, request, or command from the device.

At act **316**, the microprocessor **108** signals the control device **104** to resume sampling data as discussed above with respect to act **304**. For example, the microprocessor **108** may send a signal to the control device **104** to re-execute the process **300**. Because the microprocessor **108** has stored the converted data in a non-volatile format, the control device **104** is free to sample additional data, which may overwrite the data stored in the buffer **106**. The converted data stored by the microprocessor **108** may be provided to the external storage at any time before or after the microprocessor **108** signals the control device **104** to resume sampling data. The process **300** then ends at act **318** in some examples, although in other examples, the process **300** may not end and may instead return to act **302** or act **304** once the microprocessor **108** signals the control device **104** to resume sampling data, with the stored converted data being available to be provided to external devices automatically or by request.

The converted data may be provided to external devices configured to enable operators to review determine a cause of the fault condition. For example, the external device may include a visualization system to visualize the converted data. FIG. **5** illustrates a block diagram of a visualization system **500** according to an example. The visualization system **500** includes the external storage **502** and a visualization tool **504**. The external storage **502** is configured to store the JSON file and, optionally, other information indicative of a fault condition of the electrical device **100** or other electrical devices. The visualization tool **504**, which may include one or more software programs, is configured to interpret information stored in the external storage **502**, such as the JSON file, to generate a visualization of the information.

For example, FIG. **6** illustrates a visualization **600** of information that may be generated based on the JSON file. The visualization **600** illustrates an inverter current for each phase of a three-phase power device, including a first graph **602** of inverter current in an inverter of a first phase of the three-phase power device, a second graph **604** of inverter current for a second phase of the three-phase power device, and a third graph **606** of inverter current for a third phase of the three-phase power device.

The first graph **602** includes a first PWM signal trace **608** indicative of a PWM signal provided to a first switching device of an inverter in the first phase of the three-phase power device, a second PWM signal trace **610** indicative of a PWM signal provided to a second switching device of the inverter in the first phase of the three-phase power device, a third PWM signal trace **612** indicative of a PWM signal provided to a third switching device of the inverter in the first phase of the three-phase power device, a fourth PWM signal trace **614** indicative of a PWM signal provided to a

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fourth switching device of the inverter in the first phase of the three-phase power device, and a first current trace **616** indicative of an inverter current provided by the inverter of the first phase of the three-phase power device.

The second graph **604** includes a fifth PWM signal trace **618** indicative of a PWM signal provided to a first switching device of an inverter in the second phase of the three-phase power device, a sixth PWM signal trace **620** indicative of a PWM signal provided to a second switching device of the inverter in the second phase of the three-phase power device, a seventh PWM signal trace **622** indicative of a PWM signal provided to a third switching device of the inverter in the second phase of the three-phase power device, an eighth PWM signal trace **624** indicative of a PWM signal provided to a fourth switching device of the inverter in the second phase of the three-phase power device, and a second current trace **626** indicative of an inverter current provided by the inverter of the second phase of the three-phase power device.

The third graph **606** includes a ninth PWM signal trace **626** indicative of a PWM signal provided to a first switching device of an inverter in the third phase of the three-phase power device, a tenth PWM signal trace **628** indicative of a PWM signal provided to a second switching device of the inverter in the third phase of the three-phase power device, an eleventh PWM signal trace **630** indicative of a PWM signal provided to a third switching device of the inverter in the third phase of the three-phase power device, a twelfth PWM signal trace **632** indicative of a PWM signal provided to a fourth switching device of the inverter in the third phase of the three-phase power device, and a third current trace **634** indicative of an inverter current provided by the inverter of the third phase of the three-phase power device.

Accordingly, the visualization **600** illustrates information indicative of an inverter. For example, the visualization **600** may illustrate a current through the one or more inverter switches **222** of the inverter **212**. Using the first graph **602** as an example, the first current trace **616** may indicate an inverter current received by, provided by, or internal to, the inverter **212**. Each of the PWM signal traces **608-614** may indicate a PWM signal provided by the control device **104** to a respective inverter switch of the one or more inverter switches **222**, which may include four inverter switches. In other examples, a visualization may illustrate any information based on the contents of the JSON file and/or other information stored in the external storage **502**. For example, a visualization may include any information that may aid in analyzing a cause of a fault condition.

The examples discussed above enable device information in a data vector, the device information having been accumulated by a control device, to be repeatedly acquired. In some examples, the control device **104** may acquire the device information responsive to a sample condition being met. The device information may be stored on a temporary basis. If a fault condition is detected, for example, by a surveillance circuit, the device information may be provided to a microprocessor for non-temporary storage. The device information may subsequently be analyzed, such as by enabling an operator to view a visualization of the device information, to determine a cause of the fault condition. A cause of the fault condition may therefore be determined without storing a considerably large amount of information, at least because information may be sampled only in response to a sample condition being met, and non-transiently stored only in response to a fault condition being detected, for example.

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FIG. 7 illustrates a process **700** of capturing device information based on a sample condition being met according to an example. As discussed above, the process **700** may be an example of act **304**. For purposes of explanation, examples of the process **700** are provided with respect to FIG. 1. As discussed above, the additional components **114** may include the switching devices **116**. The switching devices **116** may be configured to operate based on PWM signals received from the control device **104**. For example, the additional components **114** may include switching devices configured to operate based on PWM signals received from the control device **104**.

At act **702**, the process **700** begins.

At act **704**, the control device **104** may determine whether a PWM signal change has occurred. A PWM signal change may include a PWM signal provided by the control device **104** transitioning from a logical HIGH level to a logical LOW level, or from a logical LOW level to a logical HIGH level, or both. The signal change may be a signal change of any of the PWM signals provided by any of the **30** example switching devices in some examples. If a PWM signal change has been detected (**704 YES**), then the process **700** continues to act **706**.

At act **706**, the control device **104** captures device information and stores it in the first memory (for example, the buffer **106**) as a data vector, as discussed above with respect to act **304**. Capturing the device information at act **706** may include determining a time at which the control device **104** captures the device information. The process **700** then ends at act **708**. In examples in which the process **700** is an example of act **304**, the process **300** may then continue to act **306**.

Returning to act **704**, if the control device **104** determines that a PWM signal change has not occurred (**704 NO**), then the process **700** continues to act **710**.

At act **710**, the control device **104** determines whether a time period has elapsed since the control device **104** most recently captured device information. As discussed above with respect to act **706**, the control device **104** may determine a time at which device information is captured each time the control device **104** captures device information. Accordingly, at act **710**, the control device **104** determines whether a time between a present time and the time at which the device information was most recently captured meets or exceeds the time period. The time period may be a specified time period, such as $2\ \mu\text{s}$ or another time period. If the time period has not yet elapsed (**710 NO**), then the process **700** returns to act **704**. Otherwise, if the time period has elapsed (**710 YES**), then the process **700** continues to act **706**. As discussed above, at act **706**, the control device **104** captures device information and stores it in the first memory (for example, the buffer **106**) as a data vector. The process **700** then ends at act **708**, which may include the process **300** continuing to act **306**.

Accordingly, in this example, the control device **104** captures device information each time a PWM signal change occurs or, if no signal change occurs within a specified time period, at the end of the time period. A data vector containing the device information, such as the data vector **400**, may then be temporarily stored in a memory, such as the buffer **106**, with other data vectors including previously captured device information.

FIG. 8 illustrates a schematic view of a data vector generation scheme **800** implemented by the control device **104** to generate the data vector **400** according to an example. The control device **104** may be driven by a clock signal generated by a clock **802**, which may be included within the

control device 104 or may be coupled to the control device 104. In some examples, the clock signal has a period of 20 ns. The clock signal may be provided to a counter 804 and a delta count module 808, and may be used (for example, by the control device 104) to generate the PWM signals 806, such as by determining a period of each of the PWM signals 806. The counter 804 is configured to provide a timestamp signal when the data vector 400 is generated, the timestamp signal being indicative of, or used to generate, the timestamp information 404. The timestamp signal, which may include a counter value, may be used to determine an absolute time value. The PWM signals 806 may be generated based on the clock signal, which may provide a signal resolution of the PWM signals. The delta count module 808 is configured to determine an elapsed time period since the delta count module 808 count was most recently reset.

The delta count module 808 generates a command to store a data vector each time the delta count module 808 determines that the elapsed time period since the delta count module 808 was most recently reset has reached a specified time period, such as 2 μ s, and each time one of the PWM signals 806 changes state. Acts 704 and 710 of the process 700, above, discuss these features. The command to store the data vector 400 commands the delta count module 808 to reset a count, and also commands the control device 104 to store the data vector 400.

As discussed above, the data vector 400 includes gate signal information 402, timestamp information 404, surveillance information 406, current measurement information 408, DC voltage measurement information 410, and temperature information 412. The gate signal information 402 may be provided by the control device 104 itself, which provides the PWM signals. The timestamp information 404 may be provided by the counter 804. The surveillance information 406 may be provided by a surveillance register 810, which may be a component of, and indicate a status of, the control device 104, such as the presence or absence of a fault condition. The surveillance register 810 may store information received from the one or more surveillance circuits 118. The current measurement information 408, DC voltage measurement information 410, and temperature information 412 may be provided by the ADC 102 by sampling one or more sensors, such as the first set of one or more sensors 112 and/or the second set of one or more sensors 120. For example, the control device 104 may request that the ADC 102 to provide the current measurement information 408, DC voltage measurement information 410, and temperature information 412 responsive to the command to store the data vector 400. In another example, the ADC 102 may provide the current measurement information 408, DC voltage measurement information 410, and temperature information 412 independently of the command to store the data vector 400, but the control device 104 may use the received current measurement information 408, DC voltage measurement information 410, and temperature information 412 to generate the data vector 400 responsive to the command to store the data vector 400.

FIG. 10 illustrates a schematic view of a data vector generation scheme 1000 implemented by the control device 104 to generate the data vector 900 according to an example. The data vector generation scheme 1000 may be substantially similar to the data vector generation scheme 800. For example, the clock 1002, the counter 1004, the PWM signals 1006, the delta count module 1008, and the surveillance register 1010 may be substantially similar to the clock 802, the counter 804, the PWM signals 806, the delta count module 808, and the surveillance register 810, respectively.

The data vector generation scheme 100 may further include ADC data 1012, which may be substantially similar to the ADC data 812, except that the ADC data 1012 is used to generate the voltage measurement information 910 which, as discussed above, may differ from the DC voltage measurement information 410. The ADC data 1012, and the data vector generation scheme 1000 as a whole, may otherwise be substantially similar to the ADC data 412 and data vector generation scheme 800, respectively.

In some examples, the data vector 400 and/or the data vector 900 may include additional, less, and/or different information than that indicated in the examples above. Each portion of the data vector 400 and/or data vector 900 may have a different length than in the examples provided above. Although certain memory and storage components have been referred to herein, it is to be appreciated that the memory and storage components may be implemented with alternate technologies than those specifically mentioned. Furthermore, in some examples, examples of volatile memory may instead be implemented by non-volatile memory components.

As discussed above, act 710 may include determining whether a time period has elapsed since device information was most recently acquired. In some examples, act 710 may alternately or additionally include determining whether a time period has elapsed since the process 700 began. For example, when the control device 104 executes the process 700 for the first time, the control device 104 has not previously acquired device information and thus may be unable to determine whether a time period has elapsed since any device information was most recently acquired. Determining whether a time period has elapsed since the process 700 began may thus serve a substantially similar purpose as determining whether a time period has elapsed since device information was most recently acquired.

Although certain examples may include three-phase power modules, examples discussed herein may be implemented in connection with different devices, including single-phase power modules and devices other than power modules. Information acquired and/or stored by an FPGA, or a device operating in a similar manner as the control device 104 discussed above, such as may sample information responsive to one of two sample conditions, or a different number of sample conditions, which may be the same conditions as those discussed above or may include different, fewer, or additional conditions.

As discussed above, in some examples a fault may occur in a power device. For example, a fault may occur in a switch of a power converter in a power device. The principles of the disclosure may be applied to acquire fault information for the fault in the power device. In other examples, the principles of the disclosure may be applied to acquire fault information for a fault outside of the power device (for example, upstream of the power device or downstream of the power device).

For example, as discussed above, the power device 200 may receive power at the input 202. The power device 200 may receive power from a power supply, such as a utility mains, generator, and so forth. If a fault occurs in the power supply, the power received at the input 202 may be disrupted or distorted. Sensors configured to sense information indicative of the power received at the input 202 (for example, the sensors 112 and/or 120), such as voltage and/or current sensors, may acquire data indicative of the fault. Accordingly, the principles of the disclosure may be applied to faults occurring outside of the power device 200, within the power device 200, or a combination of both.

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The principles of the disclosure may therefore be applied to detecting faults connected to any of various power devices. In some examples, the principles of the disclosure may be applied in connection with any electrical device configured to receive power from a power supply, and may be applied to fault detection for that produce effects that may be sensed by the electrical device even if the fault is not occurring in a component within the electrical device.

For example, the principles of the disclosure may be applied to a transfer switching system. FIG. 11 illustrates a block diagram of a power system 1100 including a transfer switching system 1102 according to an example. The power system 1100 includes the transfer switching system 1102, an arbitrary number power supplies 1104 (“power supplies 1104”), and at least one load 1106 (“load 1106”). The transfer switching system 1102 includes processing circuitry 1108 and routing circuitry 1110.

The transfer switching system 1102 is coupled to each of the power supplies 1104 at a respective input, and is coupled to the load 1106 at an output. The transfer switching system 1102 is configured to provide power derived from at least one of the power supplies 1104 to the load 1106. For example, the transfer switching system 1102 may default to providing output power derived from a first power supply 1104a of the power supplies 1104. As long as acceptable power is available from the first power supply 1104a, the transfer switching system 1102 may provide a conductive path between the first power supply 1104a and the load 1106. The routing circuitry 1110 may include one or more switching devices to route power from a desired input to the output.

As discussed below, the processing circuitry 1108 may include fault-detection circuitry to detect parameters indicative of a fault in the received power. For example, the processing circuitry 1108 may monitor the power received from the first power supply 1104a to determine if the power is acceptable. If the power becomes unacceptable, the processing circuitry 1108 may determine that the first power supply 1104a may be experiencing a fault. The transfer switching system 1102 may then disconnect the first power supply 1104a from the load 1106, and connect a second power supply 1104b of the power supplies 1104 to the load 1106 (for example, simultaneously with, or immediately before or after, disconnecting the first power supply 1104a). For example, the routing circuitry 1110 may include one or more switching devices to route power between the load 1106 and one or more of the power supplies 1104, which may be controllable by the processing circuitry 1108. The second power supply 1104b may then provide power to the load 1106 via the transfer switching system 1102.

Accordingly, the transfer switching system 1102 is configured to provide power derived from any of various power supplies 1104 to the load 1106. The transfer switching system 1102 may monitor power received from the power supplies 1104 and transfer between power supplies 1104 if the received power becomes unacceptable. The transfer switching system 1102 may therefore ensure that acceptable power is provided to the load 1106 even if a default power supply experiences a fault.

FIG. 12 illustrates a block diagram of the processing circuitry 1108 according to an example. The processing circuitry 1108 includes at least one control device 1200 (“control device 1200”), at least one first memory 1202 (“first memory 1202”), at least one second memory 1204 (“second memory 1204”), one or more sensors 1206 (“sensors 1206”), and at least one analog-to-digital converter 1208 (“ADC 1208”). The routing circuitry 1110 includes at

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least one input at least one input 1210 (“inputs 1210”), at least one output 1212 (“output 1212”), at least one switching device 1214 (“switching device 1214”),

The control device 1200 is coupled to the first memory 1202, the second memory 1204, the ADC 1208, and the switching device 1214. In some examples, the control device 1200 is coupled to the sensors 1206 directly in addition to, or in lieu of, being coupled to the sensors 1206 via the ADC 1208. The first memory 1202 is coupled to the control device 1200. The second memory 1204 is coupled to the control device 1200.

The inputs 1210 are coupled to the switching device 1214. In some examples, the inputs 1210 may be coupled to the power supplies 1104. For example, the inputs 1210 may include a respective input for each of the power supplies 1104, and each input may be coupled to a respective power supply. The output 1212 is coupled to the switching device 1214. In some examples, the output 1212 may be coupled to the load 1106. The switching device 1214 is coupled to the control device 1200, the inputs 1210, and the output 1212. The switching device 1214 may include one or more switches to connect any one of the inputs 1210 to the output 1212. The control device 1200 may control the switching device 1214 to establish a desired connection between any of the inputs 1210 and the output 1212.

The sensors 1206 are coupled to the ADC 1208 and, in some examples, to the control device 1200. In some examples, the sensors 1206 are coupled to one or more other components of the transfer switching system 1102. For example, the sensors 1206 may be coupled to the inputs 1210, the output 1212, the switching device 1214, and/or other components of the transfer switching system 1102. In various examples, the sensors 1206 may be coupled to the inputs 1210, the switching device 1214, and/or a conductor coupled between the inputs 1210 and the switching device 1214 to sense a voltage and/or current of power provided by the power supplies 1104. In some examples, the sensors 1206 may be coupled to the output 1212, the switching device 1214, and/or a conductor coupled between the output 1212 and the switching device 1214 to sense a voltage and/or current of power provided to the load 1106.

The ADC 1208 is coupled to the control device 1200 and the sensors 1206. The ADC 1208 may be configured to receive sensed information from the sensors 1206, which may include one or more analog signals. The ADC 1208 may convert the sensed information to one or more digital signals and provide the signals to the control device 1200.

In some examples, the processing circuitry 1108 may include a microcontroller which includes the control device 1200 and the first memory 1202. The control device 1200 may include a processor, such as a central processing unit. The first memory 1202 may include a FIFO buffer, such as an SRAM device. The second memory 1204 may include non-volatile memory, such as a Flash or EEPROM device. In other examples, the processing circuitry 1108 may include additional or different components.

FIG. 13 illustrates a process 1300 of operating the transfer switching system 1102 according to an example. In some examples, the control device 1200 may execute the process 1300. The process 1300 may be executed at least while power is received at the inputs 1210. In some examples, the power received at the inputs 1210 may be provided to the output 1212 via the switching device 1214, as discussed above.

At act 1302, the control device 1200 receives sensor data. The control device 1200 may receive the sensor data from the sensors 1206, directly and/or via the ADC 1208. The

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sensors **1206** may acquire the sensor data by sensing parameters of the transfer switching system **1102**. For example, as discussed above, the sensors **1206** may sense voltage and/or current parameters indicative of input power received from the inputs **1210**, output power provided at the output **1208**, other power internal to the processing circuitry **1108** and/or routing circuitry **1110**, and so forth. The sensors **1206** may acquire the sensor data responsive to a sample condition being met, as discussed in greater detail below.

At act **1304**, the control device **1200** stores the sensor data in the first memory **1202**. The first memory **1202** may include one or more buffers and may be substantially similar or identical to the buffer **106**. The first memory **1202** may include a FIFO buffer configured to store data vectors encoding the sensor data. For example, act **1304** may include the control device **1200** storing a data vector, which may be similar to the data vector **400**, in a substantially similar manner as act **304**.

At act **1306**, the control device **1200** determines whether the sensor data is indicative of the presence of a fault. The fault may be occurring in electrical-device components of at least one of the power supplies **1104** that provides the input power to the input **1206**. However, the fault may cause a disruption or distortion in the input power. The control device **1200** may detect these abnormalities in the input power and determine, based on the abnormalities, that a fault has been detected. For example, the control device **1200** may determine that a fault is present if a voltage, current, or frequency (for example, of voltage and/or current) of the input power falls outside of an acceptable range (for example, by being below a threshold value), if certain harmonics are present or absent or fall outside of an acceptable range, if a temperature of the transfer switching system **1102** falls outside of an acceptable range, a combination thereof, and so forth. In some examples, the control device **1200** may be communicatively coupled to an external device configured to detect a fault and send an indication of the fault to the control device **1200**. For example, the power supplies **1104** may provide an indication of a fault to the control device **1200**. Accordingly, in some examples, act **1306** may include the control device **1200** receiving an indication of a fault, and the control device **1200** may or may not analyze the sensor data to determine the presence of a fault.

If a fault has not been detected (**1306 NO**), then the process **1300** returns to act **1302**. Acts **1302-1306** may thereafter be repeated. As new sensor data received at act **1302** is stored in the first memory **1202** at act **1304**, old sensor data may be pushed out of the first memory **1202**. Accordingly, the first memory **1202** may store data on a temporary basis until the control device **1200** determines that a fault has been detected (**1306 YES**), responsive to which the process **1300** continues to act **1308**.

At act **1308**, the control device **1200** stores the contents of the first memory **1202** in the second memory **1204**. In various examples, the second memory **1204** is non-volatile memory configured to store information on a non-temporary basis. As discussed above, the contents of the first memory **1202** may be useful inasmuch as the sensor data stored therein, acquired up to and concurrently with the fault, may be used to analyze the fault. Accordingly, by transferring the contents of the first memory **1202** to the second memory **1204** for non-temporary storage, the sensor data may be stored for analysis at a later point in time.

As discussed above, the sensors **1206** may acquire sensor data responsive to a sample condition being met. A sample condition may include any of various conditions, such as a

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period of time having elapsed, a zero-crossing of input power received at the input **1206** or provided at the output **1212**, a combination thereof, and so forth.

The fault detected at act **1306**—YES may include a fault that prevents an active power supply from providing acceptable power to the inputs **1210**. Accordingly, the control device **1200** may control the switching device **1214** to disconnect the input coupled to the power supply experiencing the fault from the output **1212** responsive to detecting the fault (**1306 YES**), and to connect a new input coupled to a power supply not experiencing a fault to the output **1212**.

In various examples, the control device **1200** may store the sensor data in the second memory **1204** at act **1308**. The control device **1200** may store the sensor data in a data vector including the sensor data and, in some examples, additional information. For example, the data vector may include packet length or checksum data, a fault identification number, a timestamp, a fault-type identifier, the sensor data, a combination thereof, and so forth.

As discussed above, sensor data may include any of various types of sensor data that might be used to detect and/or diagnose a fault. For example, sensor data may be indicative of a voltage provided by each of the power supplies **1104**, a current provided by each of the power supplies **1104**, a bank current of each of the power supplies **1104**, a fuse voltage of each of the power supplies **1104**, and so forth. In some examples, sensor data may be acquired based on a property of the power provided by the power supplies. For example, sensor data may be acquired a certain number of times (for example, 32 times) per voltage cycle of the power supplies **1104**.

Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

What is claimed is:

1. An electrical device comprising:

a first input configured to receive first power from a first power supply;

a second input configured to receive second power from a second power supply different from the first power supply;

an output configured to provide at least one of the first power or the second power to at least one load;

a switching device configured to switchably couple at least one of the first input or the second input to the output;

a first memory configured to store data temporarily;

a second memory, the second memory being non-volatile memory; and

at least one control device configured to:

control the switching device to couple the first input to the output to provide the first power to the at least one load,

acquire data indicative of at least one parameter of the first power responsive to a sample condition being met,

store the acquired data in the first memory,

determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components,

store the acquired data in the second memory responsive to determining that the fault condition exists, and

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control the switching device to couple the second input to the output to provide the second power to the at least one load.

2. The electrical device of claim 1, wherein the first memory is a buffer.

3. The electrical device of claim 2, wherein the buffer is a first-in, first-out buffer.

4. The electrical device of claim 1, wherein the second memory includes ferroelectric random-access memory.

5. The electrical device of claim 1, wherein the sample condition includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the first power was most recently acquired by the at least one control device.

6. The electrical device of claim 1, further comprising a microprocessor configured to provide the acquired data to a visualization system to analyze the fault condition.

7. The electrical device of claim 1, wherein the fault condition includes a peak current being exceeded.

8. The electrical device of claim 1, wherein the fault condition includes a desaturation condition being detected.

9. The electrical device of claim 1, wherein the acquired data indicative of the first power includes at least one of current data, voltage data, or temperature data.

10. The electrical device of claim 9, wherein the acquired data includes the current data, and wherein the current data is indicative of an input current received at the first input.

11. The electrical device of claim 9, wherein the acquired data includes the voltage data, and wherein the voltage data is indicative of an input voltage received at the first input.

12. The electrical device of claim 1, wherein the fault condition is indicative of a fault upstream of the first input.

13. The electrical device of claim 1, wherein determining that the fault condition exists includes determining that a frequency of voltage at the first input is outside of a threshold range.

14. A non-transitory computer-readable medium storing thereon sequences of computer-executable instructions for operating an electrical device having a first input configured to receive first power from a first power supply, a second input configured to receive second power from a second power supply different from the first power supply, an output configured to provide at least one of the first power or the second power to at least one load, a switching device configured to switchably couple at least one of the first input or the second input to the output, a first memory configured to store data temporarily, and a second memory, the second memory being non-volatile memory, and the sequences of computer-executable instructions including instructions that instruct at least one device to:

control the switching device to couple the first input to the output to provide the first power to the at least one load; acquire data indicative of at least one parameter of the first power responsive to a sample condition being met; store the acquired data in the first memory; determine that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components;

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store the acquired data in the second memory responsive to determining that the fault condition exists; and control the switching device to couple the second input to the output to provide the second power to the at least one load.

15. The non-transitory computer-readable medium of claim 14, wherein the sample condition includes a threshold period of time elapsing since previously acquired data indicative of the at least one parameter of the first power was most recently acquired.

16. The non-transitory computer-readable medium of claim 14, wherein determining that the fault condition exists includes determining that a frequency of voltage at the first input is outside of a threshold range.

17. The non-transitory computer-readable medium of claim 14, wherein determining that the fault condition exists includes determining that a peak current is being exceeded.

18. The non-transitory computer-readable medium of claim 17, wherein determining that the peak current is being exceeded includes determining that a current received at the first input exceeds a current threshold.

19. The non-transitory computer-readable medium of claim 14, wherein determining that the fault condition exists includes determining that a voltage is outside of a threshold range.

20. The non-transitory computer-readable medium of claim 19, wherein determining that the voltage is outside of the threshold range includes determining that an input voltage at the first input is outside of the threshold range.

21. A method for operating an electrical device having a first input configured to receive first power from a first power supply, a second input configured to receive second power from a second power supply different from the first power supply, an output configured to provide at least one of the first power or the second power to at least one load, a switching device configured to switchably couple at least one of the first input or the second input to the output, a first memory configured to store data temporarily, and a second memory, the second memory being non-volatile memory, the method comprising:

controlling the switching device to couple the first input to the output to provide the first power to the at least one load;

acquiring data indicative of at least one parameter of the first power responsive to a sample condition being met; storing the acquired data in the first memory;

determining that a fault condition exists based on the acquired data, the fault condition being indicative of a fault of one or more electrical-device components;

storing the acquired data in the second memory responsive to determining that the fault condition exists; and

controlling the switching device to couple the second input to the output to provide the second power to the at least one load.

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